

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**CO2\_AT2 — VISUALIZATION DEVELOPMENT EXERCISE**

|                  |                                                                                                                                          |               |                                               |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------|
| Course Code:     | DSA06                                                                                                                                    | Course Title: | Data Handling and Visualization               |
| CO Assessed:     | CO2 – Development (BL4)                                                                                                                  | Assessment:   | Assessment Tool 2 – Visualization Development |
| Weightage:       | 20% of CIA                                                                                                                               | Total Marks:  | 25 (5 Questions × 5 Marks)                    |
| CO2 Statement:   | <i>Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight.(BL4)</i> |               |                                               |
| Student Name:    | Sania Herbert                                                                                                                            | Reg. No.:     | 192324062                                     |
| Section / Batch: | 2023/AI&DS                                                                                                                               | Date:         | 06.08.26                                      |

**Code of Conduct**

*I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sania

**Instructions**

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

**Annexure A – VISUALIZATION DEVELOPMENT**

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Rubrics for Calculation **ASSESSMENT RUBRIC**  
**AT2 — Visualization Development Exercise**

*CO2 · BT4: Analyze ·*

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

| Criteria                         | Weightage  | Level 4 – Exemplary                                                                                                                                                         | Level 3 – Proficient                                                                                          | Level 2 – Developing                                                                                                              | Level 1 – Beginning                                                                                                                |
|----------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Understanding of Concepts</b> | <b>25%</b> | Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight. | Good understanding of chart combination and annotation/narrative technique, with only minor gaps.             | Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them. | Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told. |
| <b>Explanation</b>               | <b>25%</b> | The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.                              | The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.   | The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.                          | Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.                                        |
| <b>Application</b>               | <b>20%</b> | Effectively applies multi-chart coordination, annotation, and storytelling techniques to produce a                                                                          | Applies development techniques to produce a workable dashboard, with only minor errors or missed refinements. | Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.         | Fails to apply appropriate development techniques; the charts, annotation, and narrative do not work together as                   |

| Criteria            | Weightage  | Level 4 – Exemplary                                                                                                                              | Level 3 – Proficient                                                                        | Level 2 – Developing                                                                                                   | Level 1 – Beginning                                                                                    |
|---------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|                     |            | complete, cohesive dashboard for the scenario.                                                                                                   |                                                                                             |                                                                                                                        | a dashboard.                                                                                           |
| <b>Organization</b> | <b>15%</b> | Charts are clearly coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity. | Charts are organized and mostly coordinated, with only minor issues in alignment or layout. | Basic structure; charts are present but poorly aligned or the annotation/narrative feels disconnected from the layout. | Poorly organized; charts are not coordinated, and the annotation or narrative is missing or misplaced. |
| <b>Accuracy</b>     | <b>10%</b> | All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.                             | Mostly accurate, with only minor omissions or a small error in one chart or the annotation. | Partially correct; some plotted values or the annotated point are missing or incorrect.                                | Largely inaccurate; major errors in plotted values, scales, or the annotated data point.               |

### Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

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| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## Problem 21

### Nonprofit Fundraising Dashboard (5 Marks)

**Data: Donations and donor retention by channel: Online = \$150k / 60% retention, Events = \$80k / 75%,**

**Direct Mail = \$50k / 40%, Corporate = \$200k / 85%.**

**Task: Build a dashboard with a bar chart of donations and a bar chart of retention rate by channel.**

**Annotate the channel offering the best long-term value, and write a narrative recommending where to focus fundraising effort next year.**

#### Dataset:

| Channel     | Donations (\$k) | Retention (%) |
|-------------|-----------------|---------------|
| Online      | 150             | 60            |
| Events      | 80              | 75            |
| Direct Mail | 50              | 40            |
| Corporate   | 200             | 85            |

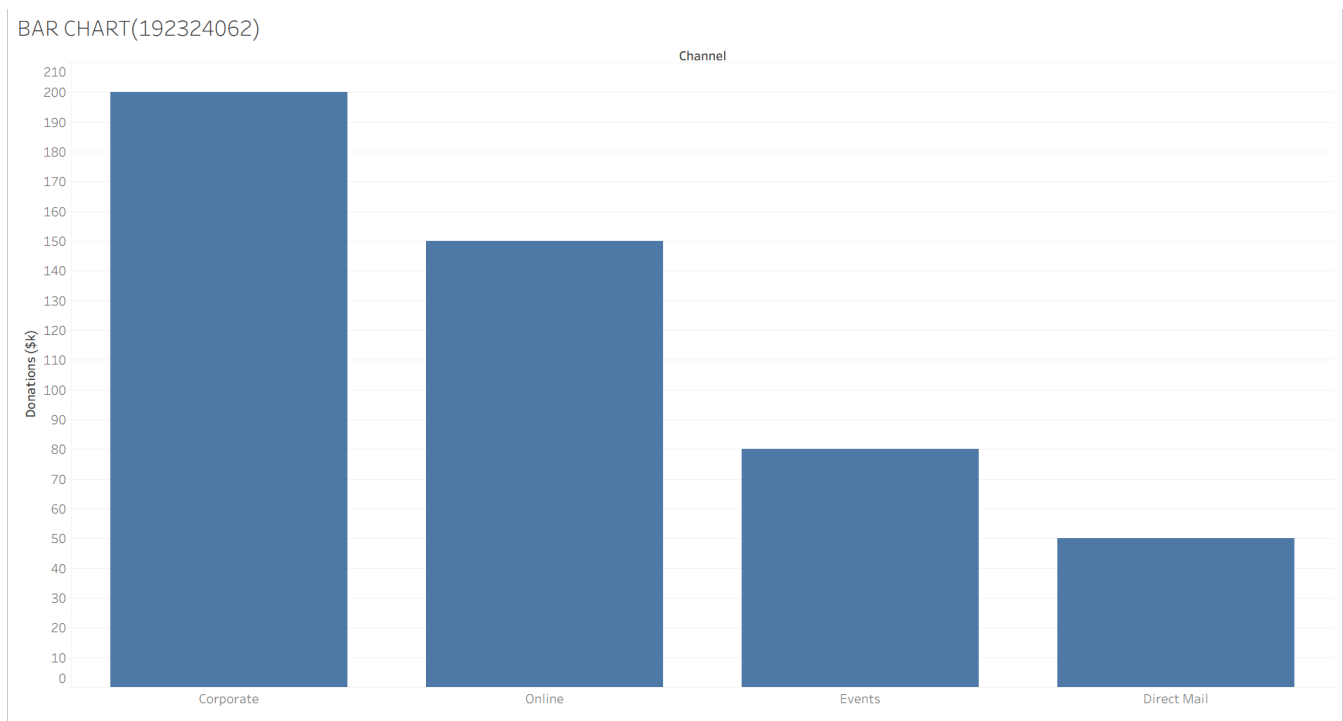
#### Chart 1

##### Bar Chart

- X-axis → Channel
- Y-axis → Donations

##### Reason

Bar charts are ideal for comparing donation amounts across different fundraising channels because the categories are discrete and comparisons are straightforward.



## Chart 2

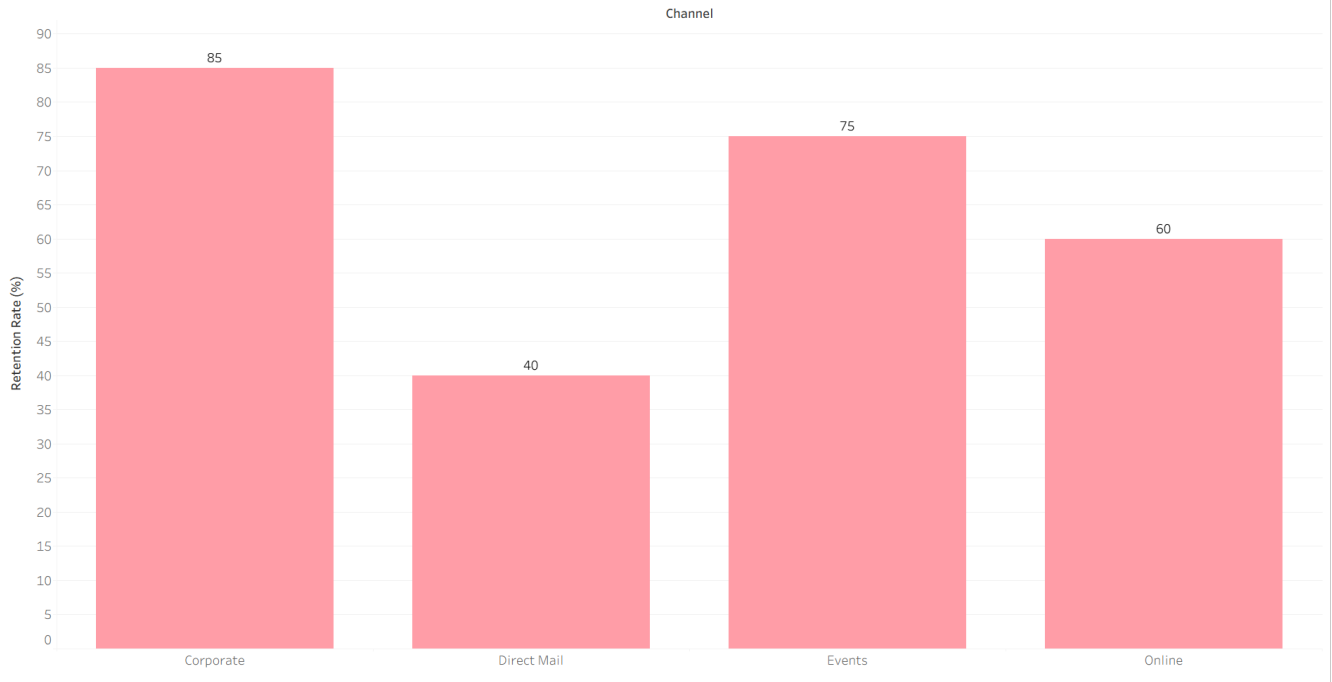
### Bar Chart

- X-axis → Channel
- Y-axis → Retention Rate

### Reason

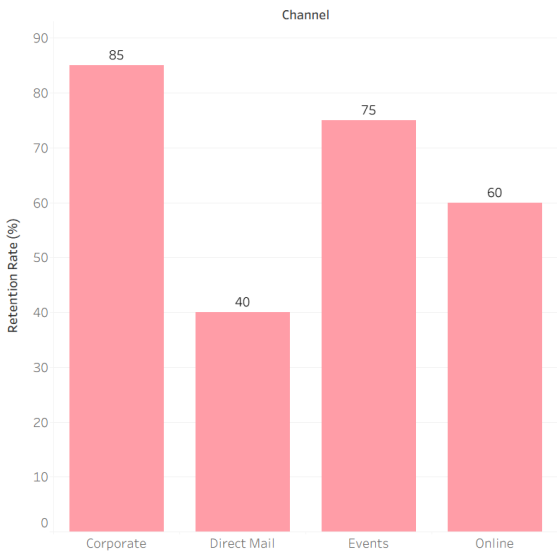
A second bar chart allows direct comparison of donor retention while keeping the same category order, making it easy to compare with donation amounts.

Retention Chart(192324062)

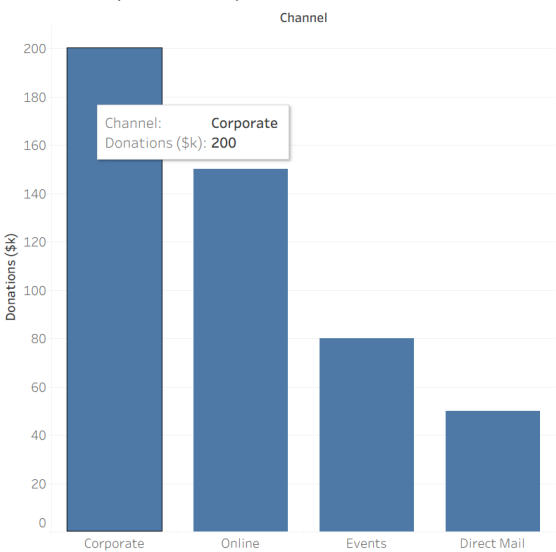


# Dashboard Layout

Retention Chart(192324062)

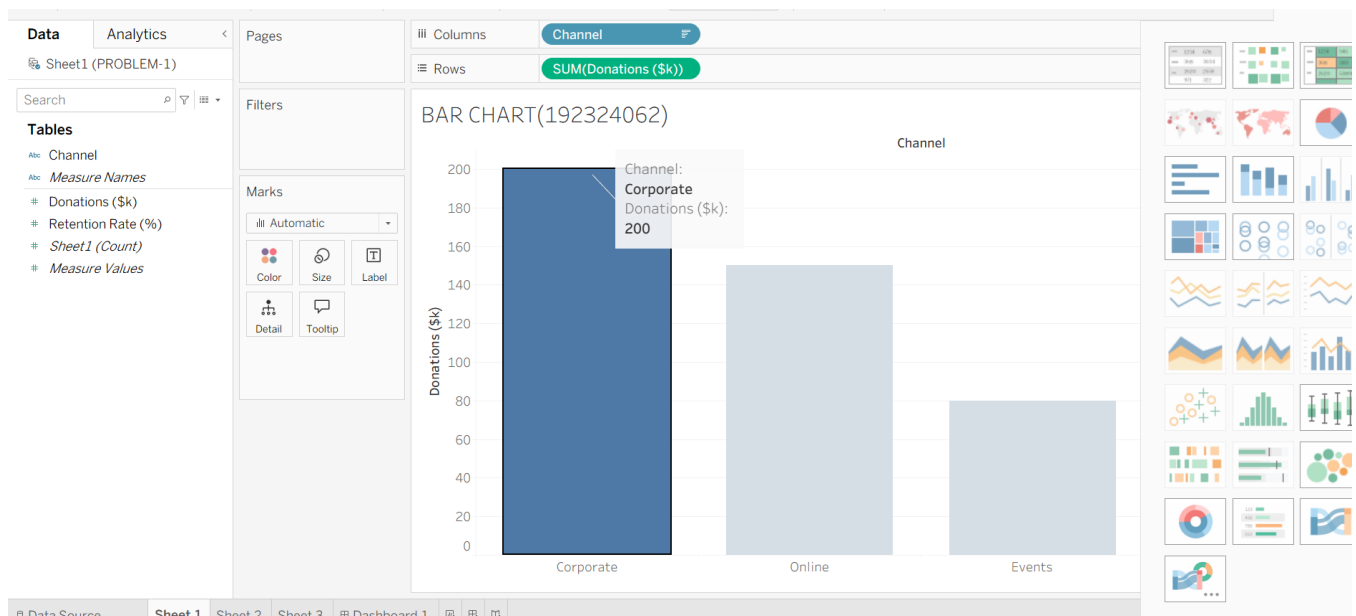


BAR CHART(192324062)



## Annotation

Annotate **Corporate**



## Narrative

Corporate fundraising generates both the highest donation amount and the strongest donor retention, making it the most sustainable fundraising channel. The nonprofit should expand corporate partnerships while continuing online campaigns to attract new donors.

## Problem 22

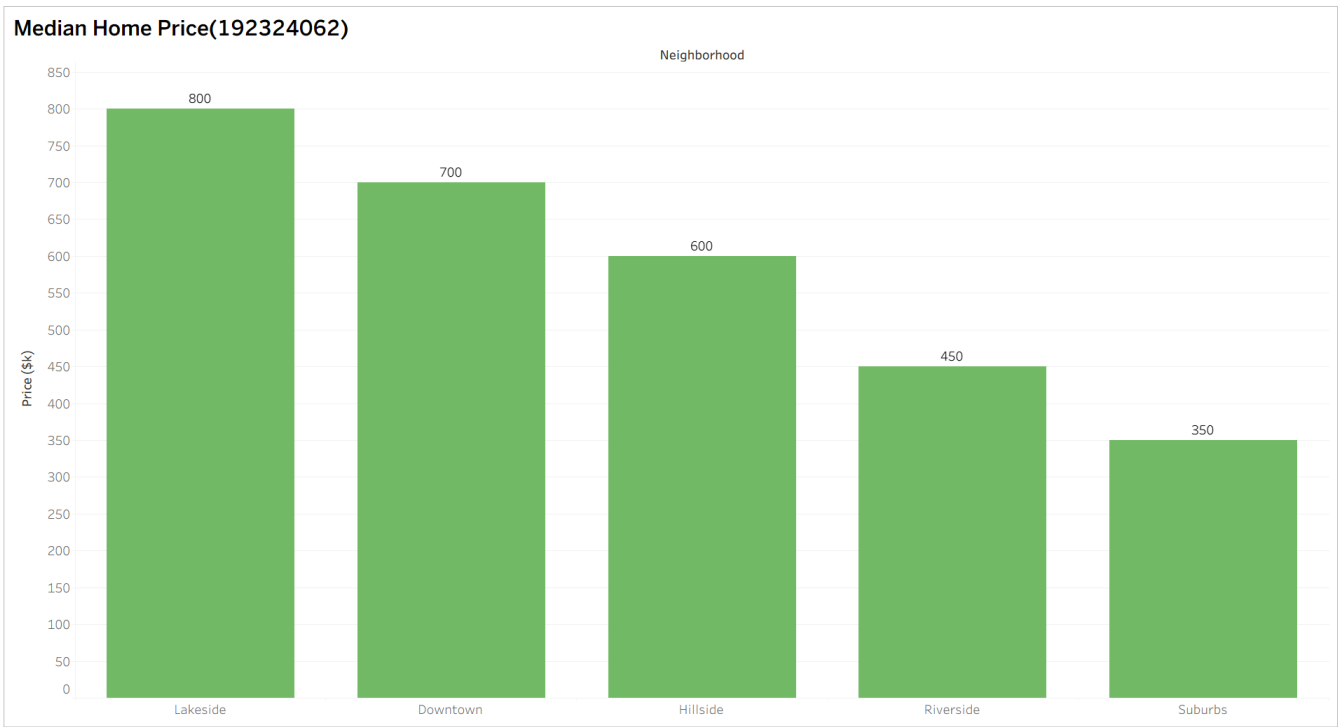
### Real Estate Market Dashboard

**Data: Median home price and days-on-market by neighborhood: Riverside = \$450k / 20 days, Hillside = \$600k / 45, Downtown = \$700k / 15, Suburbs = \$350k / 60, Lakeside = \$800k / 25.**

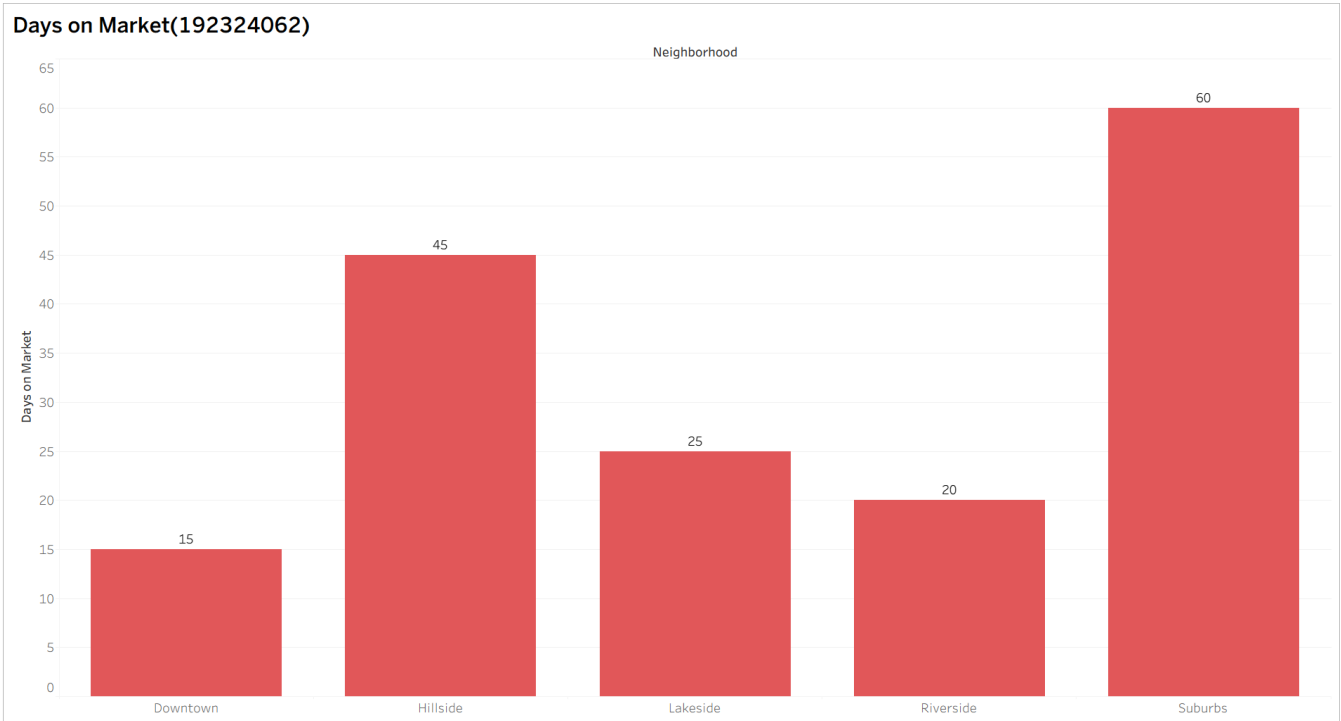
**Task: Build a dashboard with a bar chart of median price and a bar chart of days-on-market for the same neighborhoods. Annotate the neighborhood with the strongest buyer demand (low days-on-market despite high price), and write a takeaway for a home buyer.**

Dataset:

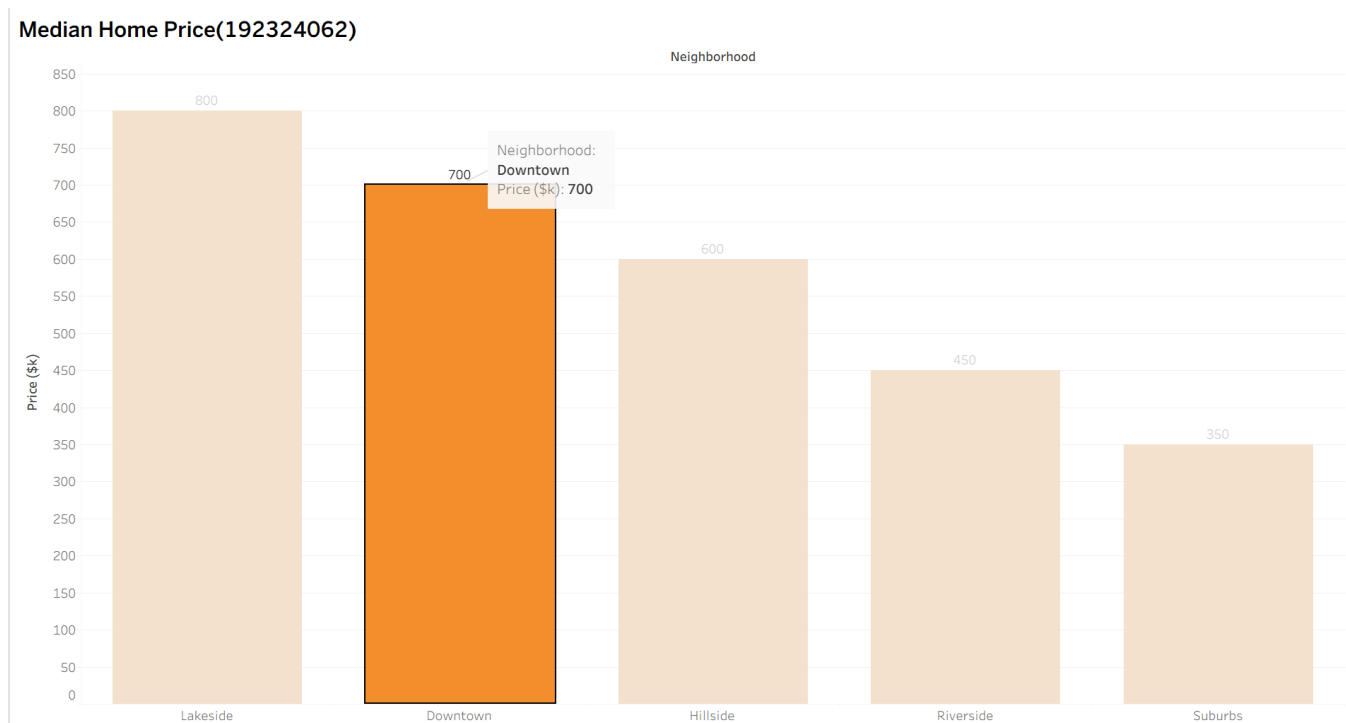
| Neighborhood | Median Price | Days on Market |
|--------------|--------------|----------------|
| Riverside    | 450          | 20             |
| Hillside     | 600          | 45             |
| Downtown     | 700          | 15             |
| Suburbs      | 350          | 60             |
| Lakeside     | 800          | 25             |







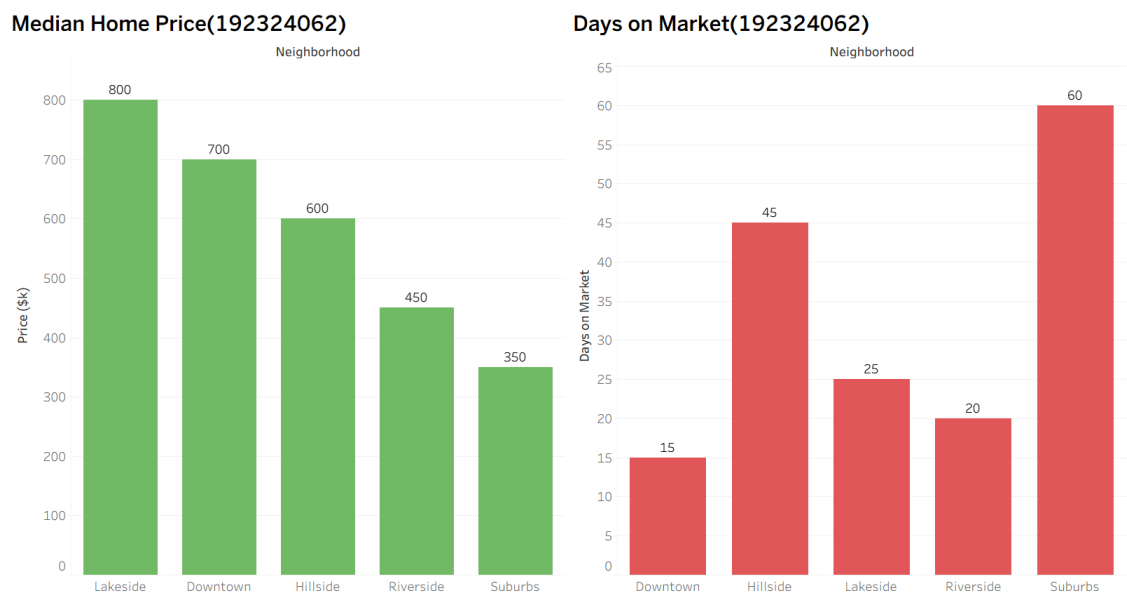
## Add Annotation



Downtown combines a high median home price with the lowest days on market,

indicating strong buyer demand. Buyers should act quickly when properties become available, while sellers can expect competitive interest

Create Dashboard



Problem 23

Fitness App Cohort Dashboard (5 Marks)

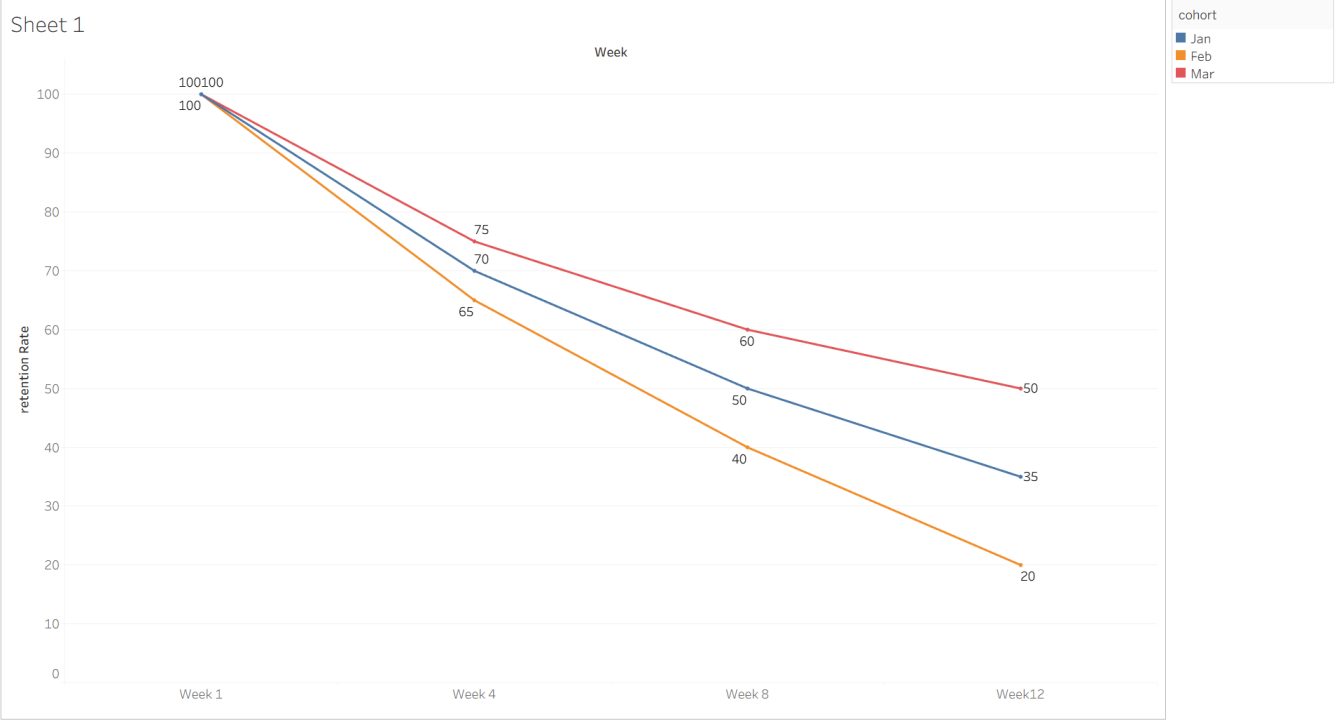
Data: Retention rate at week 1 / 4 / 8 / 12 for three signup cohorts: Jan cohort = 100%,70%,50%,35%;

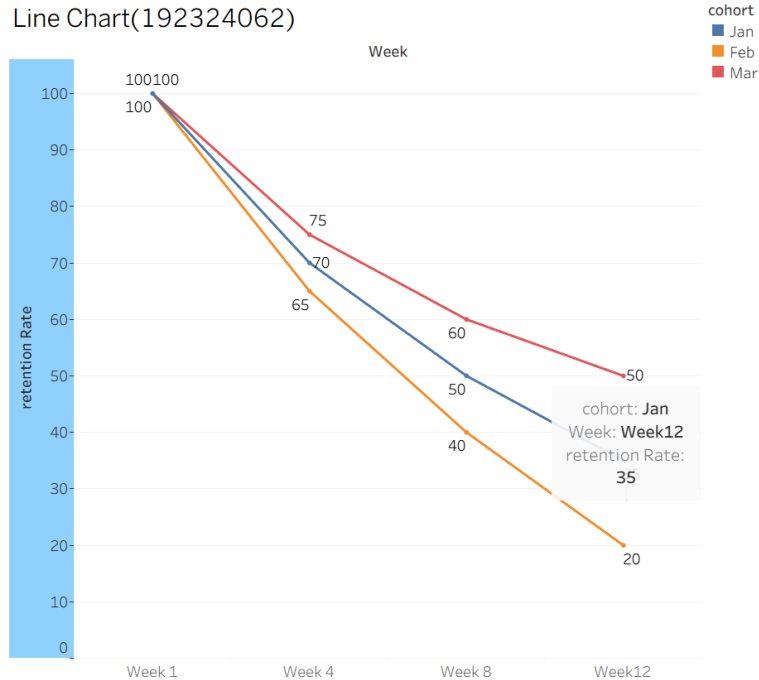
Feb cohort = 100%,65%,40%,20%; Mar cohort = 100%,75%,60%,50%.

Task: Build a dashboard with a multi-line chart showing retention curves for all three cohorts across the four checkpoints. Annotate the cohort with the steepest drop-off, and write a narrative recommending the best point to intervene.

Dataset:

| Week   | Jan | Feb | Mar |
|--------|-----|-----|-----|
| Week 1 | 100 | 100 | 100 |
| Week 4 | 70  | 65  | 75  |
| Week 8 | 50  | 40  | 60  |
| Week12 | 35  | 20  | 50  |





## Problem 24

### Storm Tracking Dashboard (5 Marks)

**Data:** Hourly wind speed (km/h) and barometric pressure (hPa) over 12 hours: wind =

40,45,55,70,90,110,105,85,60,45,35,30; pressure =  
1005,1002,998,992,985,978,980,988,995,1000,1003,1006.

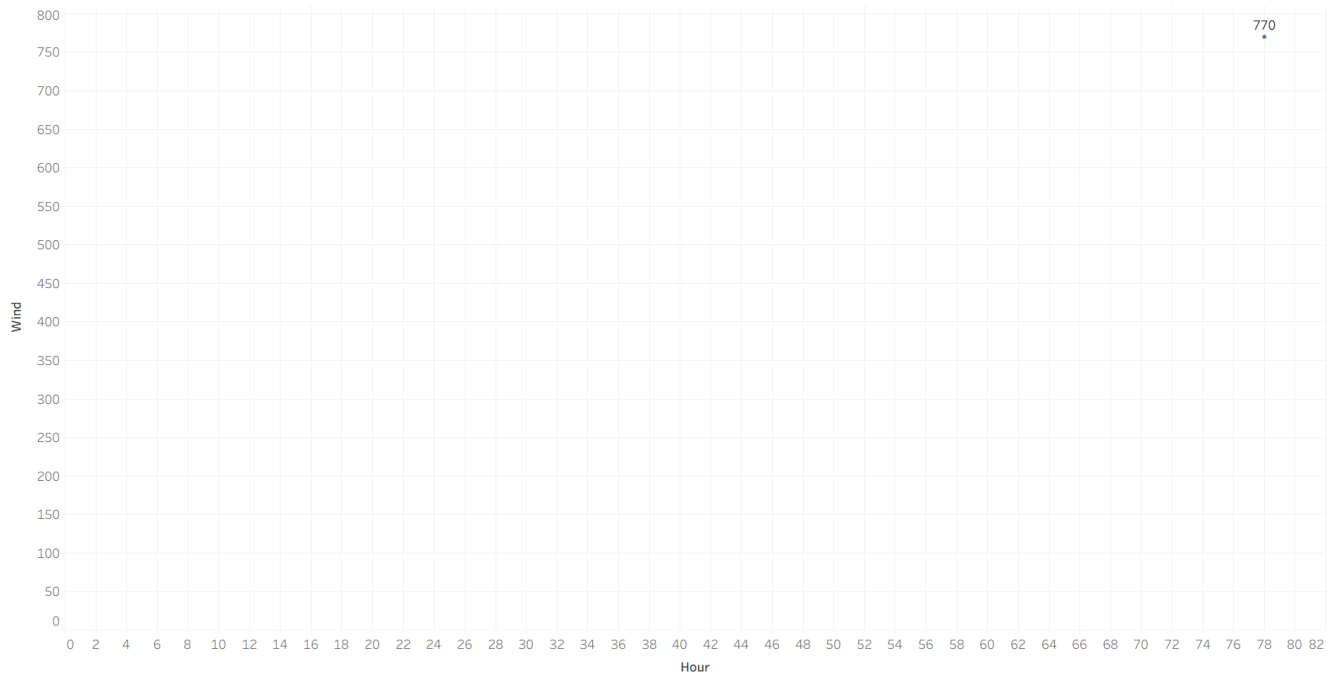
**Task:** Build a dashboard with a line chart of wind speed and a line chart of pressure on the same hourly axis, stacking the two charts vertically to avoid a dual y-axis. Annotate the peak-intensity hour, and write a narrative describing the storm's progression.

**Dataset:**

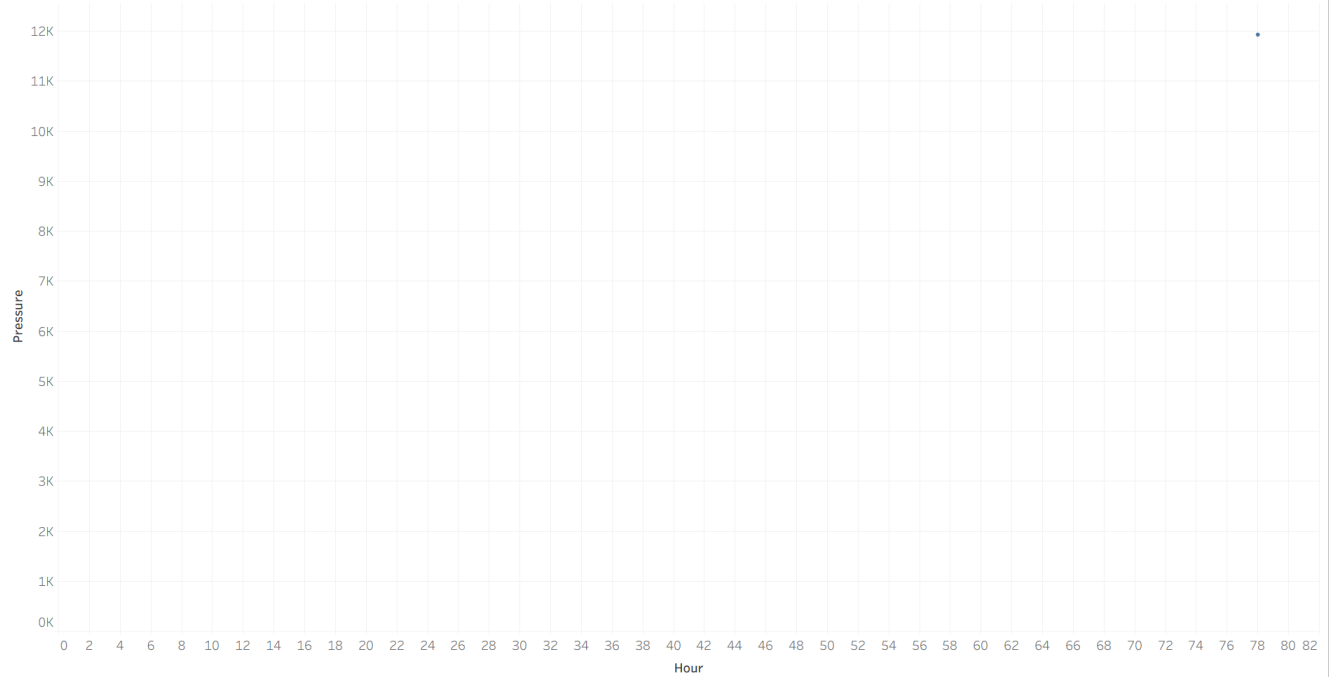
| Hour | Wind | Pressure |
|------|------|----------|
| 1    | 40   | 1005     |

| Hour | Wind | Pressure |
|------|------|----------|
| 2    | 45   | 1002     |
| 3    | 55   | 998      |
| 4    | 70   | 992      |
| 5    | 90   | 985      |
| 6    | 110  | 978      |
| 7    | 105  | 980      |
| 8    | 85   | 988      |
| 9    | 60   | 995      |
| 10   | 45   | 1000     |
| 11   | 35   | 1003     |
| 12   | 30   | 1006     |

Wind Speed(192324062)

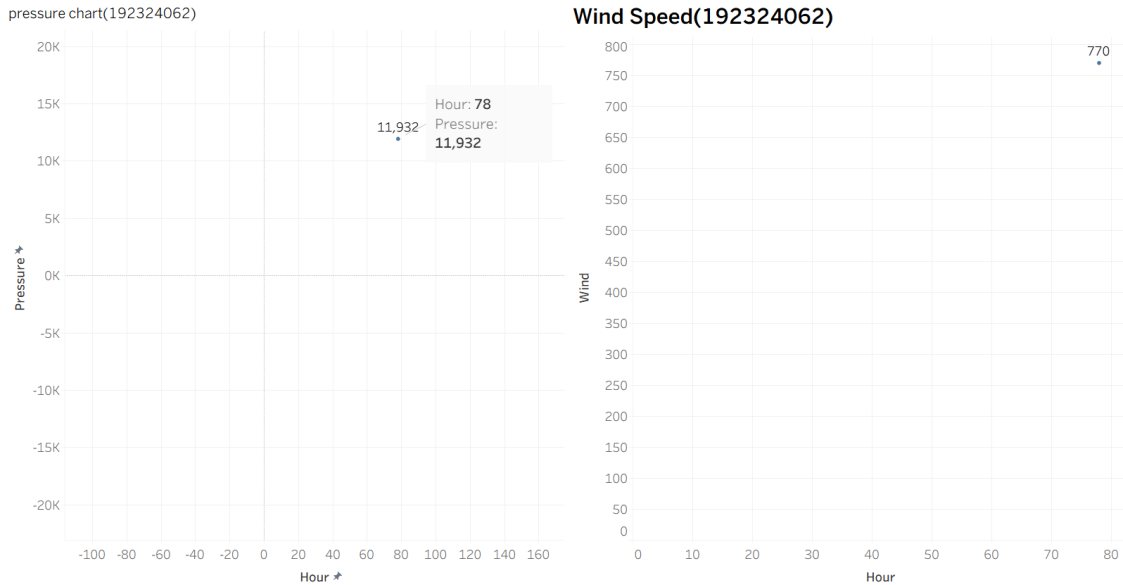


pressure chart(192324062)



pressure chart(192324062)





Problem 25

Company-Wide KPI Dashboard (5 Marks)

Data: Revenue (\$M), customer count (thousands), and employee count over 4 years: Revenue =

10,15,22,30; Customers = 50,80,120,180; Employees = 40,55,70,95.

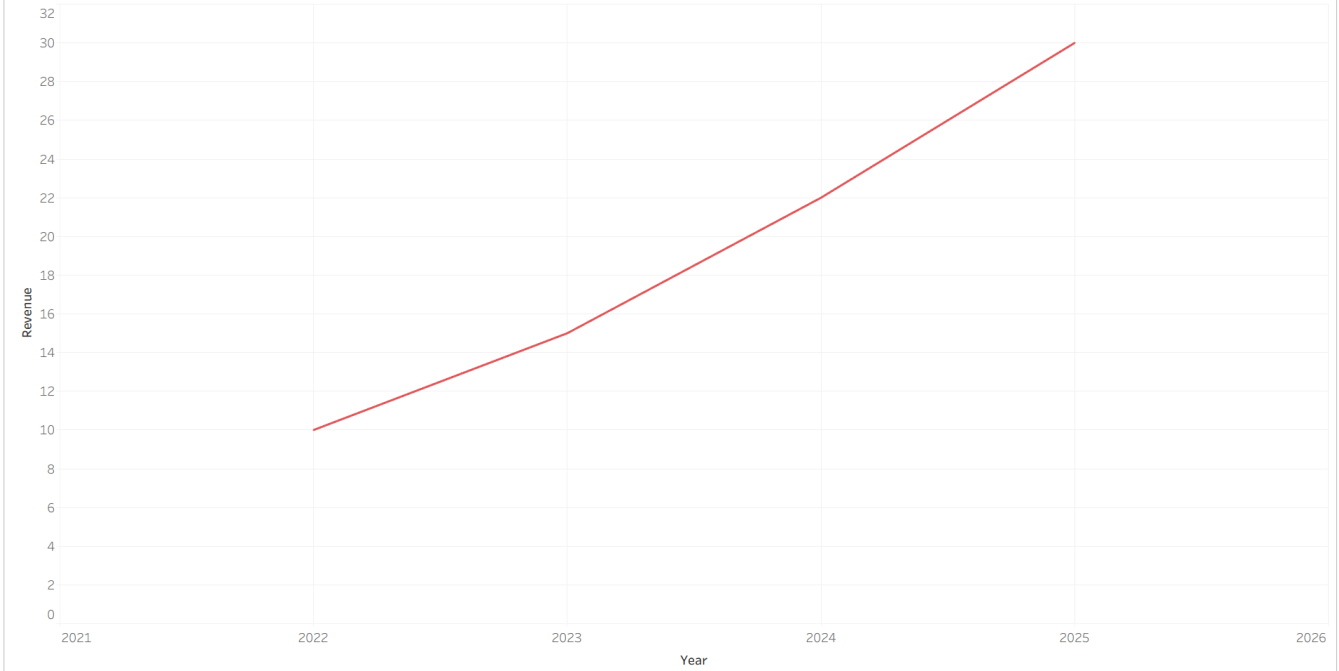
Task: Build a small-multiples dashboard with three aligned mini-charts (one per metric) sharing the same year axis. Annotate any year where growth patterns diverge across the three metrics, and write an executive-level narrative summary of the company's 4-year trajectory.

Dataset:

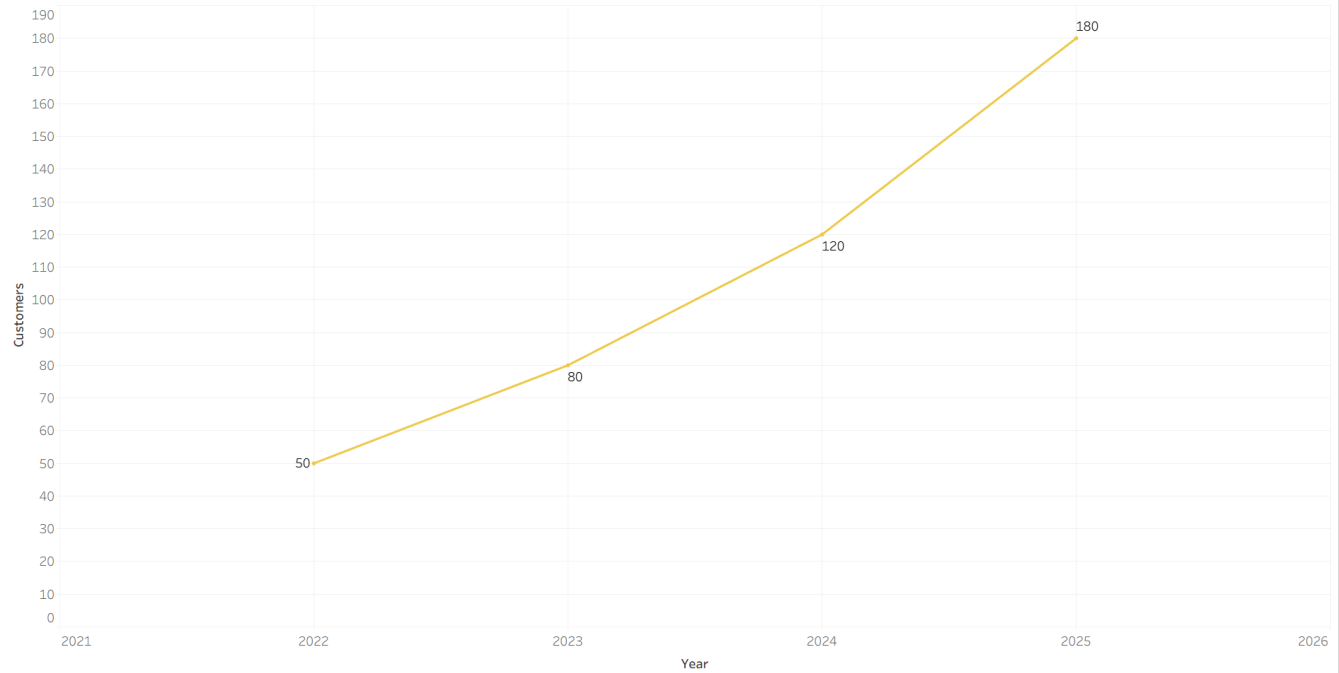
| Year | Revenue | Customers | Employees |
|------|---------|-----------|-----------|
| 2022 | 10      | 50        | 40        |
| 2023 | 15      | 80        | 55        |
| 2024 | 22      | 120       | 70        |

| Year | Revenue | Customers | Employees |
|------|---------|-----------|-----------|
| 2025 | 30      | 180       | 95        |

Revenue(192324062)

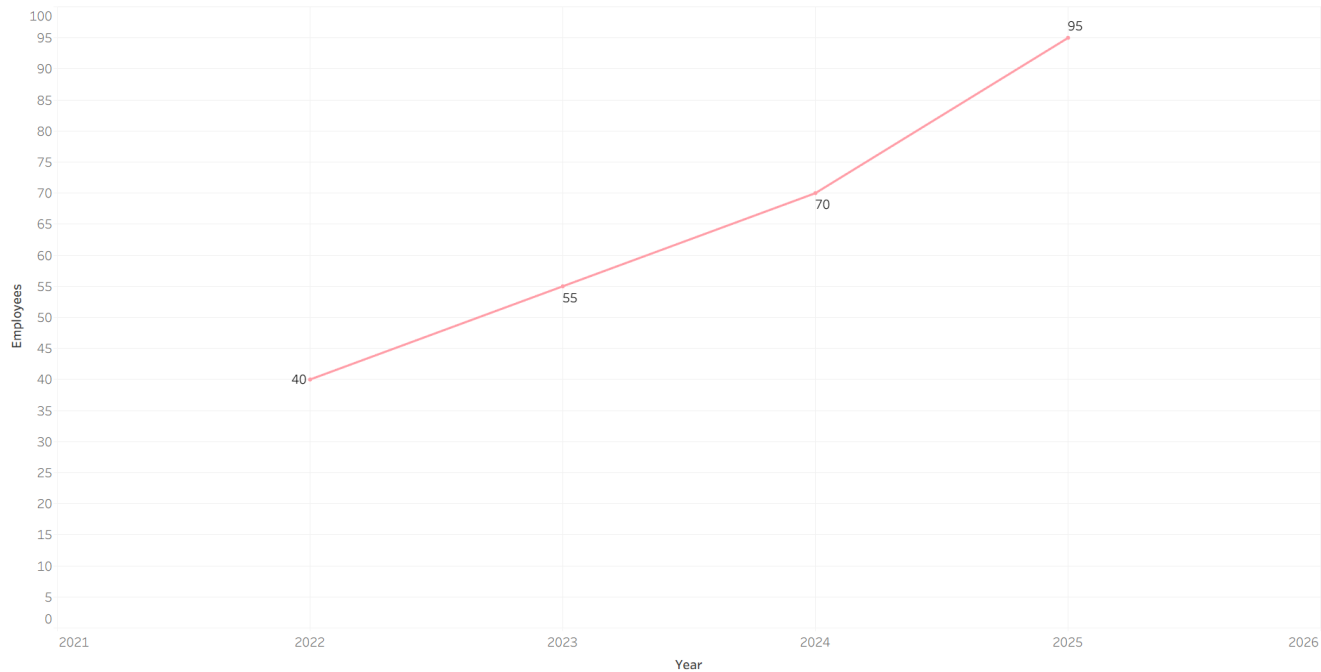


customers(192324062)

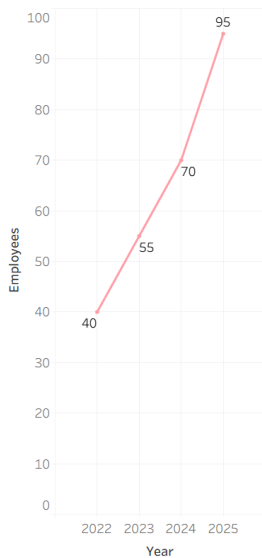




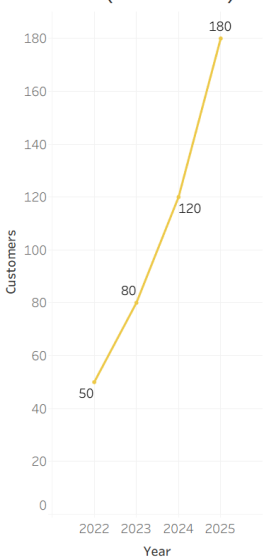
Employees(192324062)



Employees(192324062)



customers(192324062)



Revenue(192324062)

